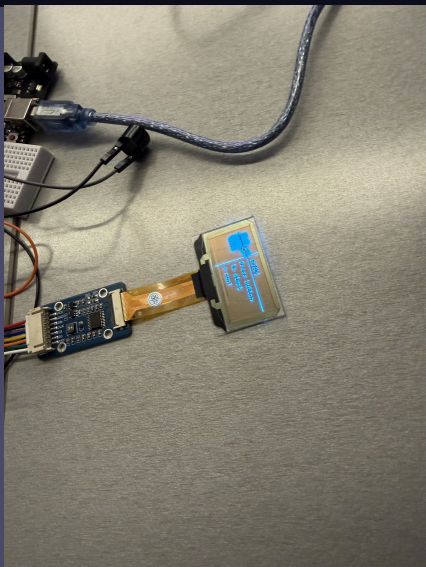


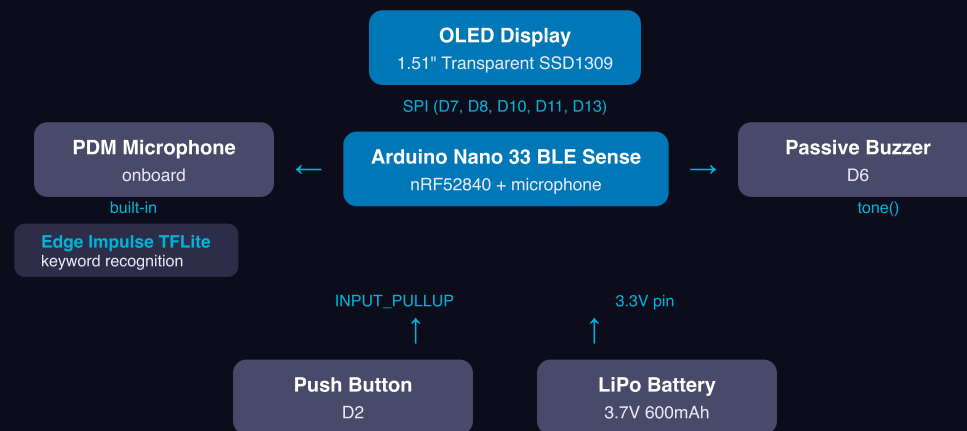


CONCEPT

RunPal is a wrist-worn running companion built around an Arduino Nano 33 BLE Sense and a 1.51" transparent OLED display. A pixel-art cat character animates in real time to reflect your running state — sitting idle while waiting, counting down, sprinting alongside you during a run, and panting on the summary screen. Runs are started and stopped entirely by voice using an on-device TFLite machine learning model trained with Edge Impulse to recognise the keywords "Hey Ajax" and "Stop run" — no phone or internet required. A passive buzzer provides audio feedback at the start and end of each run.

The system is housed in a custom 3D-printed watch case designed to fit a standard 20mm watch strap.

SYSTEM DIAGRAM



HOW TO USE

- 1 Power on**
Connect the battery or plug in via USB.
- 2 Say the command**
Say "Hey Ajax" clearly — a 5-second countdown begins.
- 3 Start running**
On GO! the timer starts and RunPal runs alongside you.
- 4 Stop your run**
Say "Stop run" or press the button — the timer freezes.
- 5 See your results**
Your run time is shown on the summary screen for 10 seconds.